COMPUTER NETWORKS 1B

CONE5112 / CONE5112w

Scenario-Based Question Bank

60 questions | 6 learning units | Questions only

Instructions to students

Apply networking concepts to each scenario and justify your reasoning. Use labelled diagrams, worked calculations and practical examples where appropriate. Show all calculation steps and state any assumptions. No answers or mark allocations are included.

Learning Unit 1: Computer Networks and Security, the Internet, Wireless and Mobile Networks

Learning Unit 2: The Application Layer

Learning Unit 3: The Transport Layer

Learning Unit 4: The Network Layer: Data Plane

Learning Unit 5: The Network Layer: Control Plane

Learning Unit 6: The Link Layer and Local Area Networks (LANs)

Original practice scenarios aligned with the learning units and outcomes in the Computer Networks 1B Module Outline 2026. All organisations and scenarios are fictional.

Computer Networks and Security, the Internet, Wireless and Mobile Networks

1.1 Connecting a school to the Internet

Scenario: A Durban school wants learners to access online lessons, send emails and upload assignments. Its principal believes that purchasing a wireless router will automatically provide Internet access.

Question: Explain the components and services the school needs to connect to the Internet. Discuss end systems, communication links, routers, an Internet service provider and network protocols.

1.2 Selecting an access network

Scenario: An accounting firm operates from an urban office with fibre coverage. Its rural branch has unreliable fixed-line connectivity, while travelling employees need access from different locations.

Question: Recommend suitable Internet access technologies for the urban office, rural branch and travelling employees. Justify your recommendations using bandwidth, reliability, mobility and deployment considerations.

1.3 Choosing a switching approach

Scenario: A training company uses its network for email, occasional file downloads and live voice communication. Its manager proposes reserving a dedicated communication circuit for every employee throughout the working day.

Question: Compare circuit switching and packet switching in this context. Evaluate which approach would use network resources more efficiently and explain the implications for live voice traffic.

1.4 Investigating network delay

Scenario: A 1,500-byte packet crosses a single link operating at 10 Mbps. The link is 2,000 km long, and signals travel at 200,000,000 metres per second. Processing delay is 2 ms and queuing delay is 5 ms. Assume no other delays.

Question: Calculate the transmission delay, propagation delay and total nodal delay. Explain how each delay arises and identify which delays are likely to vary as network traffic increases.

1.5 Explaining throughput and packet loss

Scenario: A college transfers files through three consecutive links with capacities of 100 Mbps, 20 Mbps and 50 Mbps. During busy periods, router buffers fill and packets are dropped.

Question: Determine the maximum end-to-end throughput when there is no competing traffic. Explain how the bottleneck link, competing traffic and finite router buffers affect actual throughput and packet loss.

1.6 Understanding layering and encapsulation

Scenario: A student uploads an assignment from a laptop through a wireless access point and several routers to a learning management server. The student asks why networking requires several protocol layers.

Question: Explain how the five-layer Internet model supports this upload. Trace encapsulation at the sender and decapsulation at the receiver, and explain how layered protocols enable interoperability between different manufacturers' devices.

1.7 Choosing wireless technologies

Scenario: A hospital needs wireless connectivity for staff laptops, short-range wearable sensors and ambulance crews travelling across the city.

Question: Recommend suitable wireless network types for these three requirements. Compare coverage, power requirements, mobility support and susceptibility to interference, and recommend appropriate measures to protect wireless access.

1.8 Sharing a wireless channel using CDMA

Scenario: Several monitoring devices must transmit over the same wireless frequency at the same time. An engineer proposes Code Division Multiple Access rather than assigning a separate time slot to each device.

Question: Explain how CDMA allows simultaneous transmissions using different codes. Discuss how a receiver separates the intended signal and why code selection and received signal strength affect successful communication.

1.9 Evaluating 4G LTE and 5G

Scenario: A logistics company uses mobile connectivity for vehicle tracking, video communication and warehouse sensors. Management is considering upgrading selected operations from 4G LTE to 5G.

Question: Compare the relevant architecture and capabilities of 4G LTE and 5G. Explain the roles of the radio access network and core network, and evaluate where an upgrade could benefit the company without assuming that 5G always provides better performance.

1.10 Maintaining connectivity during movement

Scenario: A consultant participates in an online meeting while travelling between cellular coverage areas. In another deployment, a mobile device must remain reachable through its home IP address while connected to a foreign network.

Question: Explain the mobility-management and handover challenges in the first situation. Evaluate how Mobile IP could support the second situation, including home addresses, care-of addresses, tunnelling and potential routing inefficiencies.

The Application Layer

2.1 Selecting an application architecture

Scenario: A college plans to distribute learning resources through either a central server or a peer-to-peer system in which students share resources directly. Some students have unreliable connections and switch their devices off frequently.

Question: Compare client-server and peer-to-peer architectures for this service. Recommend an approach based on availability, scalability, administration and resource distribution.

2.2 Identifying application requirements

Scenario: A software company is developing an online banking service, a live voice application and a background file-backup service. The development team proposes using the same transport service for all three.

Question: Analyse the reliability, delay, throughput and security requirements of each application. Explain how these requirements influence the choice of transport services and protocols.

2.3 Investigating HTTP communication

Scenario: A customer opens an online shop, requests a product page and submits an order. The browser exchanges several messages with the web server.

Question: Explain the roles of HTTP requests and responses in this interaction. Discuss appropriate methods, status codes and headers, and explain the benefits of persistent connections when retrieving a page containing multiple objects.

2.4 Improving performance with HTTP/2

Scenario: A news website contains many images, scripts and style sheets. Users report slow loading, and the developer proposes moving from HTTP/1.1 to HTTP/2.

Question: Explain how HTTP/2 multiplexing and header compression could improve performance. Evaluate why delays may still occur when the underlying TCP connection experiences packet loss.

2.5 Managing cookies and sessions

Scenario: An online retailer wants customers to remain logged in and retain shopping-cart contents as they move between pages. Customers are concerned about tracking and unauthorised account access.

Question: Explain how cookies and server-side session information can support these functions despite HTTP being stateless. Recommend measures to reduce session theft and unnecessary tracking.

2.6 Using web caching

Scenario: Students at a campus repeatedly download the same public learning resources through a limited Internet connection. The IT department proposes a shared web cache.

Question: Explain how caching could reduce response time and Internet traffic. Discuss cache hits, cache misses, freshness and conditional requests, and identify content that should not be stored in a shared cache.

2.7 Troubleshooting DNS

Scenario: Employees can reach a company web server using its IP address but cannot access it using its domain name. The problem affects some users while others can still connect normally.

Question: Explain the DNS resolution process and identify plausible causes of the different user experiences. Describe how resolver configuration, cached records and authoritative DNS information could be investigated.

2.8 Delivering adaptive video

Scenario: An online learning platform serves students whose available bandwidth changes during a lesson. A single high-quality video stream frequently pauses to buffer.

Question: Explain how HTTP streaming using Dynamic Adaptive Streaming over HTTP (DASH) could address this problem. Discuss video segments, alternative quality levels and the trade-off between playback quality and uninterrupted viewing.

2.9 Expanding through a CDN

Scenario: A South African training provider hosts all videos on one server in Johannesburg. Students in distant countries experience long start-up times, and the server becomes overloaded during examination preparation.

Question: Evaluate how a content distribution network could improve delivery. Explain content replication, server selection and the benefits and limitations of serving users from geographically distributed locations.

2.10 Developing a TCP application

Scenario: A developer must create a client-server application that allows students to submit a student number and receive a registration-status response. Multiple students may connect simultaneously.

Question: Outline the socket-programming steps required on the TCP server and client. Explain connection establishment, handling multiple clients, message boundaries and appropriate connection closure.

The Transport Layer

3.1 Understanding transport-layer responsibilities

Scenario: A learner argues that the transport layer is unnecessary because IP already delivers packets between computers. The learner also assumes that all transport protocols guarantee delivery.

Question: Use the example of a browser and an email application running simultaneously to explain transport-layer responsibilities. Distinguish process-to-process communication from host-to-host delivery and explain why reliability depends on the transport protocol used.

3.2 Delivering data to the correct application

Scenario: An employee simultaneously uses a browser, a video meeting application and an email client. Data for all three arrives through the same network interface.

Question: Explain how transport-layer multiplexing and demultiplexing deliver data to the correct applications. Discuss port numbers and how TCP connections are distinguished using source and destination addresses and ports.

3.3 Selecting UDP for sensor communication

Scenario: A temperature sensor sends a small reading every second. A late reading is usually less useful than the newest reading, and the sensor has limited processing capacity.

Question: Evaluate the suitability of UDP for this application. Describe its segment structure, connectionless operation and checksum, and explain which reliability functions would need to be implemented by the application if required.

3.4 Handling loss and corruption

Scenario: A data-transfer application operates over a link that occasionally corrupts data packets and loses acknowledgements. The first version retransmits whenever an acknowledgement does not arrive, but the receiver sometimes delivers duplicate data.

Question: Explain how checksums, acknowledgements, sequence numbers and timers can work together to provide reliable transfer. Show how the receiver can recognise and handle a duplicate packet.

3.5 Comparing pipelining protocols

Scenario: A sender transmits packets 1 to 5 using a window large enough to hold all five packets. Packet 3 is lost, while packets 4 and 5 reach the receiver.

Question: Compare how Go-Back-N and Selective Repeat would handle this event under their standard operating assumptions. Discuss receiver buffering, acknowledgements and retransmissions.

3.6 Establishing a TCP connection

Scenario: A customer opens a banking application that establishes a TCP connection before exchanging application data. A trainee suggests that one initial message should be sufficient to establish the connection.

Question: Explain the TCP three-way handshake using SYN, SYN-ACK and ACK. Discuss how the exchange establishes initial sequence information and confirms communication in both directions.

3.7 Interpreting TCP sequence numbers

Scenario: A TCP sender transmits three consecutive segments containing 500 bytes each. Their first-byte sequence numbers are 1,000, 1,500 and 2,000. The second segment is lost, while the first and third arrive.

Question: Assuming standard cumulative acknowledgements, identify the acknowledgement number after the first segment and after the out-of-order third segment. Explain how missing data can be detected and what acknowledgement follows once all 1,500 bytes have arrived.

3.8 Protecting a slow receiver

Scenario: A fast server sends data to a laptop whose application is processing received data slowly. The network has sufficient capacity, but the laptop's receive buffer is almost full.

Question: Explain how TCP flow control prevents the sender from overwhelming the laptop. Discuss the advertised receive window and distinguish flow control from congestion control.

3.9 Responding to network congestion

Scenario: A TCP connection starts transferring a large file. Its sending rate initially increases quickly, but packet loss occurs as an intermediate router becomes congested.

Question: Explain how TCP slow start, congestion avoidance and loss-recovery mechanisms regulate transmission. Discuss how a timeout may produce a different response from duplicate acknowledgements and how ECN can signal congestion.

3.10 Evaluating fairness and protocol development

Scenario: Two users share a bottleneck link. One runs a single TCP download, while the other opens several parallel TCP connections. A third application sends UDP traffic without adapting its rate.

Question: Evaluate the likely fairness of bandwidth allocation. Explain the limitations of per-connection fairness, the value of network-assisted congestion signals, and how newer approaches such as QUIC aim to address selected transport limitations.

The Network Layer: Data Plane

4.1 Distinguishing routing and forwarding

Scenario: After a link failure, a company router updates its routing information. It then sends arriving packets through a different output interface. A technician describes both activities as the same operation.

Question: Distinguish routing from forwarding using this scenario. Explain the roles of the control plane, forwarding table, input ports, switching fabric and output ports.

4.2 Applying longest-prefix matching

Scenario: A router has these forwarding entries: 192.168.0.0/16 -> A; 192.168.10.0/24 -> B; 192.168.10.128/25 -> C; and 0.0.0.0/0 -> D. Packets arrive for 192.168.10.140, 192.168.10.50, 192.168.20.5 and 8.8.8.8.

Question: Determine the output interface for each packet. Explain how longest-prefix matching resolves overlapping entries.

4.3 Evaluating traditional router control

Scenario: A business operates 40 routers. Each runs routing software and participates in distributed route calculation. Implementing a company-wide traffic policy requires changes across many devices.

Question: Explain how this traditional control-plane architecture operates. Evaluate its benefits and limitations for autonomy, convergence, administration and consistent policy implementation.

4.4 Considering software-defined networking

Scenario: A university wants to redirect selected traffic dynamically and apply consistent access rules across several buildings. Its IT manager proposes an SDN controller and programmable switches.

Question: Compare the proposed architecture with traditional networking. Explain control-plane and data-plane separation, programmability and controller availability requirements.

4.5 Understanding network service models

Scenario: A company expects its Internet connection to guarantee that every video packet arrives within 20 ms and that no packet is lost. Its provider offers a standard best-effort IP service.

Question: Explain why these expectations are not guaranteed by best-effort delivery. Compare the company's requirements with possible network-service guarantees involving delay, loss, ordering and bandwidth.

4.6 Managing router queues

Scenario: A router carries voice calls, interactive business traffic and large backups over the same congested output link. Voice quality deteriorates whenever backups run.

Question: Compare FIFO, priority scheduling and weighted fair queuing. Recommend a scheduling approach and explain the implications for voice quality, fairness, buffering and possible starvation.

4.7 Designing IPv4 subnets

Scenario: A company has the address block 192.168.40.0/24. It needs four subnets supporting at least 100, 50, 25 and 10 usable host addresses respectively.

Question: Develop a VLSM addressing plan. For each subnet, provide its prefix length, subnet mask, network address, usable host range and broadcast address, ensuring that the subnets do not overlap.

4.8 Comparing IPv4 and IPv6 operation

Scenario: An organisation is introducing IPv6. A technician assumes that IPv6 routers fragment oversized packets in the same way as IPv4 routers and that IPv6 broadcasts replace IPv4 broadcasts.

Question: Correct these assumptions. Compare relevant IPv4 and IPv6 addressing and header characteristics, including address size, fragmentation behaviour and the use of multicast instead of broadcast.

4.9 Planning an IPv6 transition

Scenario: A company has IPv4-only legacy systems, IPv6-capable new devices and a service that must be reachable by clients using either protocol. Immediate replacement of all systems is impossible.

Question: Evaluate dual stack, tunnelling and translation as transition mechanisms. Recommend where each could be appropriate and explain the operational limitations the company should consider.

4.10 Applying match-action forwarding

Scenario: A campus wants to block unauthorised traffic to a database server, prioritise approved voice traffic and direct selected traffic through a firewall. It is considering OpenFlow-capable switches.

Question: Explain how match-action rules could implement these requirements. Identify suitable match fields and actions, describe the role of OpenFlow, and distinguish switch forwarding functions from services provided by middleboxes.

The Network Layer: Control Plane

5.1 Selecting a control-plane approach

Scenario: A retail chain is expanding to several branches. Management wants consistent network policies but is concerned that a central controller failure could disrupt operations.

Question: Compare traditional distributed control with an SDN-based approach. Recommend an architecture and explain how redundancy and the distinction between logical centralisation and physical deployment affect resilience.

5.2 Calculating a shortest path

Scenario: A network contains bidirectional links with these costs: A-B = 2, A-C = 5, B-C = 1, B-D = 4 and C-D = 1. Router A must determine the lowest-cost routes to the other routers.

Question: Apply Dijkstra's algorithm from A, showing each iteration. Identify the resulting shortest paths and explain what information a link-state router requires.

5.3 Investigating a distance-vector problem

Scenario: After a destination network fails, two neighbouring routers continue telling each other that they have a route to it. Their advertised costs increase with successive updates.

Question: Explain how this situation can occur in distance-vector routing. Discuss routing loops, count-to-infinity, and the benefits and limitations of techniques such as poisoned reverse.

5.4 Using OSPF within an organisation

Scenario: A university has multiple campus networks and redundant router links. Static routing has become difficult to maintain, and link failures require manual intervention.

Question: Explain how OSPF could improve internal routing. Discuss neighbour relationships, link-state advertisements, route calculation, convergence and the use of areas as the network grows.

5.5 Connecting through two Internet providers

Scenario: A company connects to two Internet service providers and wants to exchange reachability information with both. Its administrator suggests using OSPF directly as its inter-domain routing solution.

Question: Explain why BGP is appropriate for inter-domain routing and distinguish its role from OSPF. Discuss autonomous systems, advertised prefixes and the importance of routing policy.

5.6 Applying routing policy

Scenario: An Internet provider learns two routes to the same destination. One passes through a customer and has a longer AS path; the other passes through an upstream provider and has a shorter AS path.

Question: Explain why the provider might prefer the customer route. Discuss the influence of business relationships, local preference, AS-path information and route-export policy on BGP decisions.

5.7 Managing public address advertisements

Scenario: A company receives 203.0.113.0/24 for a training exercise. An administrator mistakenly advertises a larger prefix that also covers addresses outside the assigned block.

Question: Explain the potential effects of this advertisement on other networks. Recommend address-management, prefix-filtering and route-validation practices that would help prevent incorrect or unauthorised announcements.

5.8 Deploying an anycast service

Scenario: A DNS provider runs servers in Johannesburg, London and Singapore. It wants users to reach an available instance through one shared service IP address.

Question: Explain how IP anycast could support this design. Compare anycast with unicast and multicast, and discuss why the selected server is determined by routing rather than guaranteed geographical proximity.

5.9 Improving network management

Scenario: A network team manually checks device interfaces and changes configurations one router at a time. It wants automated monitoring and more structured configuration management.

Question: Explain how SNMP, NETCONF and YANG could support these goals. Distinguish monitoring information from configuration operations, and discuss secure access, structured data and configuration validation.

5.10 Diagnosing connectivity with ICMP

Scenario: An employee cannot reach a remote server. A ping request receives no reply, while traceroute shows several responding routers followed by timeouts. The employee concludes that the server must be switched off.

Question: Explain how ICMP supports ping, traceroute and error reporting. Evaluate alternative explanations for the observations and explain why unanswered probes do not conclusively prove that a device is unavailable.

The Link Layer and Local Area Networks (LANs)

6.1 Understanding link-layer responsibilities

Scenario: A file travels from a laptop over Wi-Fi, through an Ethernet campus network and across a provider link. A learner assumes that the same link-layer frame travels unchanged from source to destination.

Question: Explain the link layer's role on each individual link. Discuss framing, medium access and error detection, and explain why link-layer technologies and frames can change along the end-to-end path.

6.2 Following data across a router

Scenario: A computer sends an IP packet to a server on another network through its default gateway. The first link uses Ethernet, and the router's next link uses a different link-layer technology.

Question: Trace encapsulation and processing at the computer, router and destination. Explain which addressing information is used for local delivery and which information supports delivery across networks.

6.3 Comparing error-detection methods

Scenario: A factory network operates near equipment that occasionally introduces transmission errors. Its technicians are comparing parity checks, checksums and cyclic redundancy checks.

Question: Compare these error-detection methods, including their ability to detect different error patterns. Explain why detecting an error does not automatically mean that the receiver can correct it or recover the missing data.

6.4 Applying parity checking

Scenario: A monitoring device uses even parity. It transmits the seven data bits 1011001 followed by a parity bit. During transmission, one data bit changes. On another occasion, two data bits change.

Question: Determine the correct parity bit for the original data. Explain how the receiver applies even-parity checking and evaluate whether the one-bit and two-bit errors will be detected.

6.5 Selecting a multiple-access protocol

Scenario: Three systems share communication media: a predictable sensor network, a network carrying occasional bursts from many devices, and an industrial network requiring orderly access.

Question: Compare channel-partitioning, random-access and take-turns protocols for these situations. Recommend an approach for each and justify your choices using utilisation, delay, overhead and collision considerations.

6.6 Explaining ARP and gateway selection

Scenario: Computer A has IP address 192.168.1.10/24 and must send data first to 192.168.1.20, then to 192.168.2.20. Its default gateway is 192.168.1.1, and its ARP cache is initially empty.

Question: Explain how Computer A determines whether each destination is local. Describe which device's MAC address it resolves through ARP for each transmission and identify the destination MAC address used in the outgoing Ethernet frame.

6.7 Investigating Ethernet switching

Scenario: A new Ethernet switch has an empty forwarding table. Computer A sends a frame to Computer B, and Computer B replies. Both computers connect to different ports on the switch.

Question: Explain how the switch handles the initial frame and the reply. Discuss source-MAC learning, unknown-unicast flooding, selective forwarding and forwarding-table ageing.

6.8 Evaluating a hub replacement

Scenario: A small office uses a shared, half-duplex Ethernet hub. Users experience poor performance when several computers transmit. The office plans to replace the hub with a switch using full-duplex links.

Question: Explain how the replacement changes collision domains and simultaneous communication. Discuss why CSMA/CD is relevant to the old shared network but is not required on correctly operating full-duplex switched Ethernet links.

6.9 Designing VLANs and a data centre network

Scenario: A college hosts finance systems, student laboratories and public web applications on a shared switching infrastructure. It wants stronger separation and better performance between application servers.

Question: Recommend a VLAN design and explain access ports, trunk links and inter-VLAN routing. Discuss how a modern data centre architecture, such as leaf-spine, could support server communication, and explain why VLAN separation still requires appropriate access-control policies.

6.10 Connecting a new host to a website

Scenario: A student connects a laptop to the campus LAN and opens a website by entering its domain name. The laptop initially has no configured IP address, DNS information or cached MAC addresses.

Question: Explain the sequence of supporting services that enables access. Include DHCP address allocation, relevant use of UDP, DNS resolution, ARP, communication with the default gateway and the eventual web connection.